**DATASCI 406 FA 26, Quiz 1.2**

**Name and Uniqname:**

## Question 1 (1 point)

You are estimating $\theta = \int_0^1 e^u \, du = e - 1 \approx 1.718$ with **antithetic variables**, starting from `tmp <- runif(5000)`.

Which line computes the antithetic estimate? **Circle one.**

1. `mean(exp(tmp) + exp(1 - tmp))`
2. `mean((exp(tmp) + exp(1 - tmp)) / 2)`
3. `mean(exp(tmp) * exp(1 - tmp))`
4. `mean(exp(tmp)) - mean(exp(1 - tmp))`

### Solution

**Answer: 2.** Each pair contributes the average of its two values, $[t(U_i) + t(1 - U_i)]/2$, and the estimate is the mean of those 5,000 pair averages. Option 1 omits the division by 2 and returns about $2\theta = 3.44$. Option 3 returns $e \approx 2.718$ every time, since $e^u e^{1-u} = e$ for any $u$. Option 4 subtracts two estimates of the same quantity and returns about 0.

## Question 2 (1 point)

All of the techniques covered in this most recent video improved the **efficiency of our estimates**. Which of the following is **not** a feature of a more efficient Monte Carlo estimator? **Circle one.**

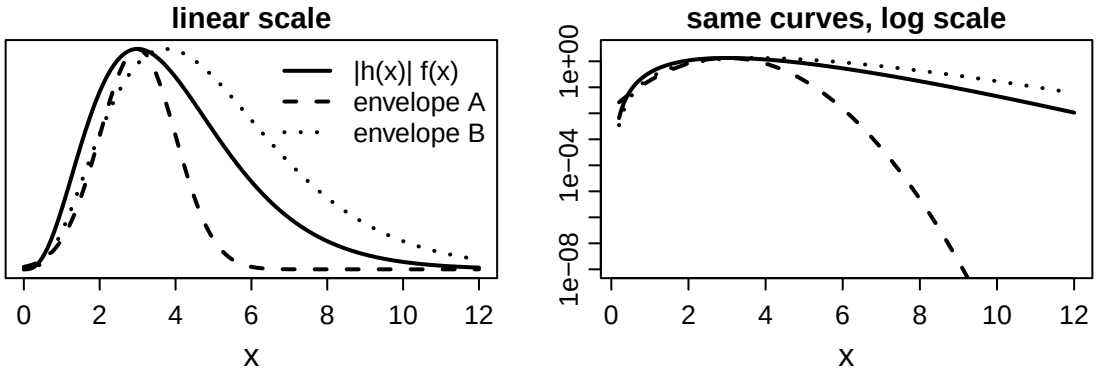
1. Narrow confidence intervals
2. Smaller variance
3. Faster convergence than $O(n^{-1/2})$
4. Asymptotic Normal distribution

### Solution

**Answer: 3.** Variance reduction shrinks the constant in front of $n^{-1/2}$; it does not change the rate. The other three do describe a more efficient estimator: the variance is smaller, so the intervals are narrower, and the estimator is still a sample mean, so it is still asymptotically normal.

## Question 3 (3 points)

You want to estimate $\theta = \mathrm{E}\left(X^3\right)$ for $X \sim \mathrm{Exp}(1)$, so $h(x) = x^3$ and $f(x) = e^{-x}$. The **solid** curve is $|h(x)|f(x)$; the **dashed** and **dotted** curves are two candidate envelope densities, A and B, each rescaled to a comparable height — so read their **shapes**, not their heights. The right panel is the same three curves on a log vertical axis, where a rescaling only shifts a curve up or down.




Circle the envelope you would use and justify the choice in two or three sentences. Your answer must say **both** (i) what the better envelope's shape has to do with $|h(x)|f(x)$, and (ii) what happens to the values $h(x)f(x)/g(x)$ for the **other** envelope when $x$ is large and why is this a problem.

Circle: **A** or **B**

**Solution**

**Answer: B.**

1. **Why B.** The variance-minimizing envelope is proportional to $|h(x)|f(x)$, so we want the candidate whose shape follows the solid curve. B does so across the whole range; A matches only near the peak.
2. **What goes wrong with A.** A falls off much faster than the target in the right tail, which the log panel shows as A bending away from the solid curve. The weights $h(x)f(x)/g(x)$ then grow without bound, so the estimate comes to depend on a few huge weights attached to draws that A almost never makes. B stays roughly parallel to the target and a little above it, so its weights stay bounded and the variance is finite.

For your information, A is $N(3, 1)$ and B is Gamma(4, rate 0.8). The variance of a single term is about 766 for A and 6.4 for B, and sampling directly from Exp(1) gives 778, so A buys essentially nothing despite sitting on the peak. B's weight works out to $6 \cdot 0.8^{-4} e^{-0.2x} \leq 14.6$. Gamma(4, rate 1) would have been exactly proportional to $x^3 e^{-x}$, giving every weight the value $\Gamma(4) = 6$ and zero variance, but choosing it requires already knowing $\theta = 6$.

**Rubric.** 1 pt for B, 1 pt for (i), 1 pt for (ii).